

## RIBEIRAO PRETO, Brazil

WMO: 836520

Lat: 21.134S

Lon: 47.774W

Elev: 1802

StdP: 13.76

Time Zone: -3.00 (SAB)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 6             | 49.0       | 52.1 | 37.8                          | 35.6 | 77.2 | 41.1 | 40.5 | 73.5 | 16.2                  | 80.4 | 14.0 | 78.6 | 4.1                   | 120  | 0.305 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 19.5                   | 96.9            | 70.1 | 95.0 | 70.5 | 93.2 | 71.0 | 78.4                | 88.1 | 77.3 | 87.4 | 76.3 | 86.2 | 7.6                  | 320  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 75.6                            | 143.5 | 81.8 | 75.0 | 140.2 | 81.4 | 73.6 | 133.7 | 80.2 | 43.5          | 88.1 | 42.2 | 87.0 | 41.2 | 86.5 | 84.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 15.5          | 13.4 | 11.8 | 43.5                       | 101.1 | 3.8                | 2.1 | 40.8                                               | 102.6 | 38.6       | 103.9 | 36.4       | 105.1 | 33.7       | 106.7 |    |    |
| (5)               |      |      | 42.1                       | 78.9  | 3.2                | 1.9 | 39.8                                               | 80.3  | 37.9       | 81.4  | 36.1       | 82.5  | 33.7       | 83.9  |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 75.6   | 79.0  | 79.7  | 78.7  | 76.1  | 71.0  | 69.3  | 69.3  | 72.5  | 76.4  | 78.8  | 78.2  | 79.0 |
|                                                                                  | DBStd       | 5.38   | 3.02  | 2.72  | 2.90  | 3.74  | 4.02  | 3.78  | 4.63  | 4.18  | 4.74  | 4.26  | 3.44  | 2.83 |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD65       | 45     | 0     | 0     | 0     | 1     | 6     | 11    | 19    | 6     | 2     | 0     | 0     | 0    |
|                                                                                  | CDD50       | 9358   | 899   | 832   | 889   | 784   | 651   | 579   | 598   | 697   | 791   | 892   | 846   | 900  |
|                                                                                  | CDD65       | 3926   | 434   | 412   | 424   | 334   | 192   | 141   | 152   | 237   | 342   | 427   | 396   | 435  |
|                                                                                  | CDH74       | 36196  | 3526  | 3601  | 3506  | 2902  | 1632  | 1335  | 1659  | 2728  | 3775  | 4376  | 3530  | 3626 |
| CDH80                                                                            | 15951       | 1541   | 1650  | 1528  | 1234  | 527   | 372   | 552   | 1217  | 1969  | 2228  | 1572  | 1561  |      |
| Wind                                                                             | WSAvg       | 5.3    | 5.3   | 4.8   | 4.7   | 4.8   | 4.7   | 4.6   | 5.2   | 5.8   | 6.1   | 6.3   | 6.2   | 5.6  |
| Precipitation                                                                    | PrecAvg     | 51.1   | 9.9   | 9.0   | 6.1   | 2.2   | 2.1   | 1.0   | 0.5   | 0.6   | 2.0   | 3.9   | 5.9   | 8.2  |
|                                                                                  | PrecMax     | 59.1   | 23.0  | 17.0  | 11.2  | 6.1   | 4.5   | 5.9   | 1.5   | 3.4   | 4.8   | 8.5   | 11.8  | 12.2 |
|                                                                                  | PrecMin     | 31.0   | 3.1   | 3.9   | 3.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.9   | 2.0   | 4.1  |
|                                                                                  | PrecStd     | 7.5    | 4.5   | 4.1   | 2.0   | 1.6   | 1.2   | 1.7   | 0.5   | 0.9   | 1.6   | 2.0   | 2.6   | 2.2  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 96.4  | 96.7  | 94.8  | 93.5  | 89.7  | 88.2  | 89.8  | 95.1  | 98.9  | 100.8 | 95.3  | 95.1 |
|                                                                                  |             | MCWB   | 71.6  | 74.6  | 75.5  | 72.4  | 69.6  | 67.4  | 67.5  | 66.2  | 68.3  | 69.2  | 73.4  | 74.0 |
|                                                                                  | 2%          | DB     | 93.3  | 93.5  | 92.8  | 91.6  | 87.7  | 86.2  | 87.8  | 91.8  | 96.7  | 97.1  | 93.3  | 92.9 |
|                                                                                  |             | MCWB   | 72.6  | 73.9  | 74.4  | 72.0  | 68.9  | 66.5  | 65.9  | 64.8  | 67.6  | 70.0  | 72.3  | 73.2 |
|                                                                                  | 5%          | DB     | 91.2  | 91.5  | 90.7  | 89.7  | 85.7  | 84.2  | 85.9  | 89.5  | 93.6  | 94.8  | 91.3  | 91.0 |
|                                                                                  |             | MCWB   | 72.6  | 73.7  | 73.6  | 72.1  | 68.1  | 66.0  | 64.7  | 64.5  | 66.7  | 69.4  | 71.8  | 72.9 |
|                                                                                  | 10%         | DB     | 88.1  | 89.5  | 88.1  | 87.6  | 82.8  | 82.2  | 83.5  | 87.4  | 91.2  | 91.4  | 89.2  | 88.1 |
|                                                                                  |             | MCWB   | 73.5  | 73.7  | 73.2  | 71.2  | 66.7  | 65.4  | 64.0  | 64.0  | 66.4  | 69.8  | 72.2  | 73.3 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 78.4  | 78.7  | 78.7  | 77.3  | 72.9  | 72.7  | 70.9  | 70.6  | 74.6  | 78.9  | 80.3  | 80.8 |
|                                                                                  |             | MCDB   | 88.0  | 90.4  | 89.7  | 88.6  | 85.0  | 82.6  | 84.1  | 87.7  | 87.4  | 87.6  | 85.9  | 85.6 |
|                                                                                  | 2%          | WB     | 76.9  | 77.3  | 77.1  | 75.5  | 71.4  | 70.2  | 68.8  | 68.2  | 72.5  | 76.4  | 77.1  | 78.2 |
|                                                                                  |             | MCDB   | 86.6  | 88.9  | 87.3  | 86.3  | 83.2  | 79.3  | 82.7  | 84.6  | 86.4  | 87.0  | 86.9  | 86.8 |
|                                                                                  | 5%          | WB     | 75.8  | 76.3  | 75.7  | 74.2  | 70.3  | 68.5  | 67.1  | 66.9  | 70.9  | 74.4  | 75.3  | 76.4 |
|                                                                                  |             | MCDB   | 84.7  | 87.2  | 85.6  | 84.5  | 80.8  | 78.5  | 80.6  | 82.6  | 85.0  | 84.9  | 85.5  | 85.2 |
|                                                                                  | 10%         | WB     | 74.7  | 75.2  | 74.6  | 72.9  | 68.7  | 67.1  | 65.5  | 65.5  | 69.4  | 72.7  | 73.8  | 75.0 |
|                                                                                  |             | MCDB   | 82.9  | 85.0  | 84.0  | 82.5  | 78.5  | 77.6  | 78.5  | 81.1  | 82.4  | 83.3  | 83.8  | 82.9 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 18.1  | 19.5  | 19.1  | 21.6  | 22.6  | 24.7  | 26.8  | 27.9  | 25.8  | 22.7  | 20.1  | 18.6 |
|                                                                                  |             | MCDBR  | 22.6  | 22.8  | 22.7  | 24.3  | 26.0  | 28.2  | 29.6  | 31.9  | 31.0  | 28.1  | 24.3  | 22.5 |
|                                                                                  |             | MCWBR  | 6.9   | 7.3   | 7.5   | 8.2   | 9.8   | 11.4  | 10.9  | 10.6  | 9.3   | 7.8   | 7.9   | 7.2  |
|                                                                                  | 5% WB       | MCDBR  | 17.9  | 20.1  | 19.9  | 21.0  | 22.2  | 23.5  | 26.1  | 26.5  | 24.1  | 22.0  | 20.4  | 18.9 |
|                                                                                  |             | MCWBR  | 7.2   | 7.6   | 7.7   | 8.2   | 9.4   | 10.9  | 10.9  | 10.3  | 9.2   | 8.7   | 8.3   | 8.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.401  | 0.402 | 0.393 | 0.373 | 0.351 | 0.353 | 0.366 | 0.407 | 0.506 | 0.468 | 0.422 | 0.410 |      |
|                                                                                  | taud        | 2.434  | 2.417 | 2.449 | 2.477 | 2.502 | 2.494 | 2.434 | 2.291 | 2.013 | 2.196 | 2.339 | 2.393 |      |
|                                                                                  | Ebn at Noon | 299    | 295   | 290   | 283   | 277   | 270   | 269   | 268   | 253   | 273   | 292   | 297   |      |
|                                                                                  | Edh at Noon | 39     | 39    | 36    | 33    | 30    | 29    | 31    | 39    | 56    | 48    | 43    | 41    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1787   | 1881  | 1718  | 1607  | 1352  | 1269  | 1355  | 1599  | 1709  | 1842  | 1849  | 1867  |      |
|                                                                                  | RadStd      | 234    | 149   | 151   | 106   | 79    | 87    | 87    | 108   | 162   | 135   | 98    | 118   |      |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page